All right. Good afternoon. Recap of the last lecture. We were talking about the properties of our maximum likelihood estimator that we had introduced last week.

So the first property that we talked about was this invariance property, which just said if we have some MLE theta hat of some parameter of interest theta, right? We're not really interested in the parameter theta itself. We want to know some function of that parameter, G of theta, right? We can get the MLE G of theta, that function or transformation by just plugging in theta hat. Theta hat is our MLE of G of Thames. So once we kind of get an MLE for theta, we get an MLE for any function of it for free. And variance property, very nice property of the maximum likelihood estimator. We've done a couple examples there. The other property we talked about was this consistency property, right, which is an asymptotic result as our sample size gets bigger and bigger, tends to infinity. What is it saying? It's saying we're basically converging to the true parameter value that says that this theta hat sub n, our MLE based on a sample size of n converges in probability to the true parameter Theta sub t as n tends to infinity. Consistency property, right? If we have infinite data, then we're basically going to the correct answer. Doesn't necessarily mean that we have a good estimate for a finite sample size. But if we did have infinite data, we'd have a good estimate in some sense. And then we talked about bias as well. We defined bias of an estimator as the expectation of that estimator, expectation of theta hat minus the true unknown value theta, right?

And if the expectation is equal to theta, we have an unbiased estimator. Otherwise, we have a biased estimate.

As we saw, the MLE is not always unbiased.

It can be, but it's not necessarily unbiased, right? But it is consistent. And so we looked at this example here, right, our uniform, zero to theta example. And we saw that the MLE that we had derived previously, the maximum X value for this parameter to theta, that MLE was biased.

This expectation, n over n plus 1 times theta, which is not equal to theta, so it's a biased estimator.

Although as N gets large, this goes to one, and the bias vanishes for very, very large samples.

So those are our properties of our maximum likelihood estimator.

Today we're going to talk about sampling distributions.

So I recall earlier in the course, we had defined a statistic.

It's just some function of the data. we take in the random data x1 x2 through x-n and we apply some function to it little t here and we spit out some value t it's a rule for taking in the data and spitting out a value it does not depend on the the population parameters theta right so once we observe the data we can calculate this and so a lot of the things we've looked at are themselves statistics very often the estimators we're using like these are examples of statistics, the MLE estimates, the Bays estimators, right? There are rules where we can take in the observed data, and we can spit out some summary or value, right, T, and so they are statistics, functions of the data. And remember, we typically think of the data X as random. It comes from some distribution. They're random variables, although eventually we hope to observe some data, in which case it will no longer be random. But we think about this more abstractly, Right, before we observe any data, X, the data itself, follows some distribution. It's random, right? And so if we transform that data, X, that is random, we end up with something else that's also random. If we took a sample X1 through XN and applied this function T, we'd get some value. If we took another sample X1 through XN and applied this function T, we'd get some other value. And we could do this again and again and again, at least in theory. and so we'd get some distribution over this random statistic T. Our goal is to kind of

understand what that distribution is, at least for certain statistics that we care about. So that's the idea. We have a statistic, a function of the random data, so it inherits that randomness. It is also random, and we want to understand the distribution of that transformed thing, that statistic. And so we call that a sampling distribution of this definition here. Consider a random sample of data. X1 through XN, from some distribution, some data distribution. And let's say that distribution has unknown parameter theta. And we have some statistic, T of X1 through XN, some function of the data. That's a statistic. Then the probability of that distribution of T, this transformation of the data, the probability distribution of that is called the sampling distribution of T of the statistic T. That's all it is. It's the probability distribution for this random function. this function of the random data. And we've said, when we calculate a statistic, it only depends on the data X. It does not depend on the population parameter of theta, but the sampling distribution itself might depend on theta. So we'll see now. We can calculate X without knowing theta, but the actual distribution that the statistic has, could depend on theta and probably will depend on theta, because it's inheriting its randomness from the underlying data distribution. So let's look at an example here. Let's say we want to estimate the proportion, P of patients who do not relapse after treatment with a drug. We have data X1 through XN. And these are IID from a Bernoulli distribution. So 1-1 data. So 1 represents no relapse. Zero represents relapse. And we have this statistic here, T. And it's just the sample proportion to the sample mean. Right? We sum up our observed X values divide by the sample size n. So it's a statistic. It's a function of the data. Given the data, we can calculate T. And we can use that to estimate P. In fact, we've already seen that's the MLE. We want to know now what's the sampling distribution. We know the formula for T.

We know how to calculate it. It's the sample proportion. But what's the sampling distribution?

Can we write down a probability mass function for this? So we know now the individual data, XI, IID from a Bernoulli distribution, parameter probability p.

Each of these end data points has this distribution. Now what if we look at the sum of the data? from i equals 1 to n of x i what's the distribution of this sum yeah back there yeah it's binomial we know if we sum up independent identical bernoulli draws we get a binomial random variable so since each x i is bernoulli we know that the sum is going to follow a binomial distribution with parameters n and p and the values that this sum can take x i can be zero one or any integer up to n can't be beyond n because we really have n trials and we can't get less than in zero successes. So these are the allowable values for this random variable, the sum of the x's. The sum of the x is, this is a random variable. It has this distribution. Well, since the sum of the x is a binomial random variable, we could write down its probability mass function. Say little f sum xI, the random variable, given the parameters n p is just the binomial probability mass function. And choose the random variable of x times p to the random variable of x is 1 minus p to power n minus the random variable, sum of x in our case. And again, the random variable sum of x and take the value 0, 1. Okay, now notice we have our statistic t equal to 1 over n, a constant, times this random variable, the sum of x. We just wrote down the probability mass function for the sum of x. Now we just have a constant times that random variable, or this is the same as saying n times t, is equal to the sum of x.

Notice the probability that this random variable t is equal to zero is what?

Yeah, probability that the sum of the x is equal to zero. It's one to one. If the sum of x is zero, t is zero. T can only be zero if the sum of x is zero. So this probability that t equals zero is the same, the probability the sum of x that's zero.

And similarly for any other value that these this random variable can take like probability over n, that's the same as the probability of the x is is one, and so on.

So it's a discrete random variable.

T is a discrete random variable, just like the sum of the x's is, and the probabilities map up perfectly. All we want is a probability mass function, or a function that tells us for any possible value of t, what the probability is, and we know those probabilities.

They're here.

They come directly from probability mass function for the sum of the x's.

So we have this. So we know a probability mass function for some possible value of the random variable t that we'll call little t given the parameters n and p is just this right but we want things in terms of t instead of the sum of the x's right but notice n times t is the sum of the x's so that tells us what the probability is for each t n choose n t or the sum of the x times p to the nt or the sum of the x times one minus p to the nt or the sum of the x's times 1 minus t or n minus the sum of the x's. And the possible values that T can take now are 0, 1 over N, 1 over N, 2 over N through 1. Okay, so since we already had the probability mass function for the sum of the xes because we recognized it as a binomial, right? And we have this perfect relationship. We kind of get this sampling distribution for T here free by plugging in, yeah. It is, yeah, the sample proportion, we call it the proportion because it's binary, but it is, yeah, this sample mean. So the central limit theorem says asymptotically, the sample distribution approaches a normal distribution, but it does not mean that the sample distribution is a normal distribution for any finite value of N. But this thing, if you plotted this for bigger and bigger N, it would look closer and closer to a normal distribution. Now, it can't actually be normal because it's discrete and the normal distribution is for continuous random variables, right? But as N gets bigger and bigger, you have more possible points, right? So it's like a closer and closer grid approximation to a normal distribution. You have more and more points of approximating that grid. If n is small here, four points, very non-normal. How can it ever look like? Yeah, it's a good question. We're going to show a plot at the end of class where we actually do this. But as N gets very, very large, right, you get less and less uncertainty. So you can imagine that normal distributions' tails are getting further and further away from the bounds as the sample size gets large. For a small sample size, certainly it could be very non-normal. It could be pushed up against the boundary, and we'll see an example of that. But as you get more and more sample size, those tails kind of shrink in, away from the bounds so they become less and less of an issue. What are you saying like T is equal to zero, one over and two over end? Because like the sum of X, it's only like discrete whole numbers, so then you're multiplying like a discrete whole number by one over N. Yeah. If sum of X is zero, T is zero. If sum of X is one, T is one over N and so on. It's just this relationship here. Some helpful properties of the sampling distribution, right? Why do we want this? Well, for one, it gives us some insight into the variability of a statistic. Is this sampling distribution very spread out? Is there a lot of uncertainty around this statistic? Is it very concentrated? Is there a very little uncertainty? And specifically when we talk about the uncertainty around a statistic, we typically use this term the standard error. All that is is the standard deviation of the sampling distribution. It's just a standard deviation, but it's the standard deviation of a specific distribution, the sampling distribution in this case. So if we're talking about the distribution of the data, in this example, the data was Bernoulli distributed. We could talk about the standard deviation of the data distribution, the standard deviation of the Bernoulli distribution. But if we start talking about the sampling distribution of this statistic T here, then if we're talking about standard deviation, we call it the standard error. Just a different term to recognize the fact that we're talking about a sampling distribution rather than the distribution of the data. And as we just kind of discussed, as the sample size N increases, the sampling distribution of many statistics, like this one, the sample mean that we looked at, becomes approximately normal, or it approaches a normal distribution asymptotically.

That's the central limit theorem. Yeah, question. Excuse me. So the sampling distribution versus the sampling distribution of the data, I thought the core distinction was the distribution of the population and the sampling distribution. But then you added a third one, what's the distinction, which is the... So there's the distribution of the data. That's what I call it. You could call it a population distribution. I call it the distribution of the data. Oh, I see. The distribution of a statistic is called the sampling distribution, or the sampling distribution of that statistic to be more specific. You might say the sampling distribution of the sample mean, for example. So it's just those two, at least in this concept. We've talked about other distributions prior and posteriors too, but in this context, there's the distribution of the data, distribution of individual X's, and then there's the distribution of a statistic, T, a function of the data, and we call that distribution the sampling distribution. The sampling distribution is not the distribution of the individual data points. It's the distribution of a function of the data. So, like, so, for example, if I go and, like, interview a bunch of people and I get, like, the mean of their responses on a scale one to 10 or whatever, that would be, like, an example of one type of sampling distribution.

Yeah. It's the sampling distribution of the sample mean in that case. If you had a different statistic like the sample, you could have the sampling distribution of the sample, you could have the sample maximum. That's an example we did last class, right? A question. So the distribution of the data, it could maybe, like, only have, like, one sample. It would be willing from one sample if that sample had many data points, but then the sampling distribution. distribution like necessarily involved like multiple or repeated sampling, right? Of like that, on some particular statistic? Yeah, it's recognizing the fact that your sample is random. And so at least in theory, you could have gotten a different sample or a different sample or a different sample. And so, you know, across all the different possible realized samples, what is the distribution of that statistic? Of course, you'll have some realized value of T because you observe a specific data set X, a specific sample. It's not talking about that. It's talking about the more abstract concept of the distribution of the statistic over all the samples you could have gotten. So just briefly, data is the entire population regardless of whether you connect and collected it. Data is the distribution of a single data point, a single X, X1, X2, they're IID, so they all have the same distribution. In this case, in this case, it's Bernoulli. The distribution of a single data point is Bernoulli. Now, a statistic is going to be a function of your whole sample, multiple Xs, right? But it's still going to output one number. It's going to collapse these N observations into one number. That number itself is a random variable, right? And the distribution of that random variable, that's a sampling distribution. So let's do a couple other examples of sampling distributions, specifically for maximum likelihood estimators, which we've been talking about that recently. Okay, so can we understand a little bit more about some of the MLEs that we've discussed? Okay, so let's talk about normal data. Let's say we have X1 through XN, IID random variables from a normal distribution. Normal distribution has parameters mean mu, variance, sigma squared. And let's say mu is unknown, but sigma squared is known. We want the sampling distribution of mu hat, which is just the MLE of the parameter. So first we need to know what is mu hat. What is the MLE of mu here? remember.

What was that? So I'm not looking for distribution. I'm just looking for, you know, if we have a normal dataset with an unknown mean, mu and a known variance, how do we calculate the MLE?

It's the sample mean. Yeah. So mu hat is just x bar, the sample mean. We sum up our X's and we divide by the samples. That's how we get the MLE for the mean in this setup. Or we could write that as one over n, the sum, xI, or we could write that as one over n times the sequence x1 plus x2 through xN or we could write this as x1 over n x2 over n through xn these are all equivalent they're all the sample mean and the sample mean is our m le in this setup we're not looking for the mLA itself we're looking for the sampling distribution of the mle the sampling distribution of x bar notice x bar is itself random because each of these x is are random if we took a new sample we'd get a different value for x

bar. If we took another sample, we'd get a different value for x bar. It's random. It's a random variable. It inherits its randomness from the randomness of the data. It must have some distribution.

We want to know what that distribution is. We know each of these X's is a normal random variable. So our MLE here is a sum, a weighted sum, of normal random variables. What do we remember from stat 131 about weighted sums of normal random variables? What family of distributions will awaited sum of normal variables? random variables be vector.

Not kye squared. If I sum two normal random variables, what distribution is that?

It's normal. Anytime we're taking a sum of normal random variables, the result is also normal. That's a property of the normal distribution specific. It's not true of any distribution. It's a property of the normal distribution.

So we know that this is going to be normally distributed. That's all we know so far. It's going to be normally distributed with some mean and some variance, But we know it's normally distributed. So if we can find the mean and the variance, we've identified the distribution, the parameters of that distribution. Sums of normal random variables, also normal. But we know mu hat here will be normal. And we just want the mean and the variance of that normal distribution. Well, we know the formula for the statistics, so let's go ahead and calculate the mean and the variance directly. The mean or the expectation, mu hat, it's just the expectation x1 over n, x2 over n, XN for n. And we remember from Stat 131 again that we have linearity of expectation. So the expectation of a sum is the sum of the expectation. And we can always pull constants outside of the expectation. So this is going to be equal to the expectation of x1 over n, the expectation of x2 over n through the expectation x sub n over n. What's the expectation of x1? Same as this one and this one and what is it? It's mu.

They're all mu. We don't know what mu is, but it's all mu. Mu for n.

And there's n of these. We have n times mu over n, which is just mu. Are we seeing the expectation of all of them is just mu because they come from the same normal distribution with the overall parameter means? Yeah, they're IID, independent, and identically distributed. So they all come from the same distribution, so they all have to have the same mean. And that mean is mu, because they're coming from a normal distribution, which is me mu. Starting how we said that it was the sample mean, is that like something we went over in the last class we were talking about, like, one over the sample mean was like the maximum likelihood estimator? Yeah, we did a normal example where we showed for normal with known variance, but unknown mean, the maximum likelihood estimate of that unknown mean is the sample mean. So we have the mean of the sampling distribution now, it's mu, same as the distribution of the data in this case.

And now we want the variance. We need both the mean and the variance. to determine which specific normal distribution this is. So the variance of mu hat now, or the variance, or n, x2, or n, x n, how can we simplify this? Can we say this is the variance of this one plus the variance of this one plus the variance of each term sequentially? It's not in general, but it is in this case. Does anybody know why it is in this case? Not because it's normal, back there? Because it's independent, yeah. If we have independence, then the variance of a sum will be the sum of the variances, but only if we have independence, right? So I'm going to say due to independence here, this is equal to the variance of x1 over n, the variance 2 over n, okay. And we know from working with variances in stat 131, if we have a constant, we pull it outside of the variance and we square it, so each of these is going to be over n squared, the variance, the random variable. And we know what each of these variances are, right? They're i.d. all come from that normal distribution with mean, mu, and variance sigma squared. So each of these has variance sigma squared. We have n terms. Each of those terms is 1 over n squared, and sigma squared. So this reduces to sigma squared over n. We've identified that the sampling distribution is going to be

normal, because it's a sum of normal random variables. And we've identified the mean and the variance of the sampling distribution. M-Hat, our M-LE, will be normally distributed, mean-mue, and variance sigma squared over n, a question. There are, we're not going to look at them, but if you're interested, there's a distribution called the Koshi distribution. It has infinite variance, and so it has that property that when you sum them, the variance isn't decreasing, or when you average, then the variance is decrease. So notice here, the sampling distribution of our M-L-E, mu-hat, our estimator for mu, has mean-m-mue. So its expectation is the true parameter. It's unbiased, and its variance is sigma squared over n. So as our sample size gets larger and larger, the variance gets smaller and smaller. We get a more precise estimate case, an MLE that's unbiased and consistent. Doesn't always have to be unbiased, though. Question. Yeah, then it would be a lot harder to derive things like the sampling distribution or maybe even writing down the likelihood function is more complex in that case. Everything in this class will be a random sample, though. so we will have IID random variables, but you don't have to necessarily. That would have to be contained in the likelihood function. You'd have to have a joint distribution for all of the random variables. Here, because of independence, the joint distribution is the product. That's why whenever we write down our likelihood, we can write it as a product, right? It simplifies to a product due to independence. But if you didn't have independence, it would be a multivariate distribution instead of a product of univariate distribution. So you would have to encode that somehow, and it would. would be in that likelihood. Wait, just to clarify, are there any other distribution than some of those right-orables get the same distribution or is it only normal? I think it's just normal. Yeah, I think that's a special property of the normal distribution. Okay, let's do one more example. Suppose we have a random sample, X1 through XN, IID from the uniform distribution on our favorite interval, zero to theta here. What's the sampling distribution of Theta hat, the maximum observed X value, or the MLE of Theta? as we've seen for other lectures. So we want the sampling distribution and the mean of that sampling distribution. Well, we kind of did this in our last class. But recall, if the random variable y is the maximum observed x value, then the probability density function for y of y, n is y to the n minus 1, the theta to the n, or y between 0 and theta. We derived that in one of our examples in the last class. our last example.

And we also calculated the expectation. I'll go ahead and do that again. So the expectation of y or the maximum observed x value, the integral from zero to theta of y, probability density function, the definition of expectation of a continuous random variable.

And we can plug in the integral from zero to theta n or theta to the end of power is y to the end of the power v.Y n, or theta to the n is 1 over n plus 1, y to the n plus 1, evaluated at the limits, theta 0, which is n where theta to the n is 1 over n plus 1 times theta plus 1, and that simplifies to n over n plus 1 times theta.

Again, we did this at the end of last class, we're just going over it again here.

So notice we actually answered both of these questions in our example last time, right? What's the sampling distribution of Theta hat? Well, that's this distribution. This is the PDF. So we have the distribution. Then it also asked what's the mean of that sampling distribution, this expectation here. And again, as we pointed out last time, if an M-O-E is consistent, as our sample size increases as N increases, our statistic, T, in this case the maximum observed X value, tends to, on average, get closer and closer to the true value theta. N over N-plus-1 term gets closer and closer to 1. expectation becomes closer and closer to the true value theta. Okay. Last piece here, so just kind of visualizing this idea of a sampling distribution, the essential limit theorem. Okay, so we looked at this example earlier where we had this proportion P coming from a Bernoulli data distribution, and we were calculating the sample proportion, this statistic T, as our MLE of that parameter P. So what I've done here is for some different sample sizes, so I considered sample size 5, 10, 30,

and 50, and different true values for the population parameter P, so considering value 3, 5.

0.1, 0.3, 5. I'm showing the PMF, the probability mass function of sampling distribution of T, right?

This is what we derived earlier. This is the true probability mass function.

And then I'm plotting a normal approximation based on the central limit.

A normal distribution centered around the same area.

And that's, in each case, this blue curve. It's continuous. And then these bars with these red dots represent the probability mass function.

Probability mass function is discrete. That's why we have bars with dots. Normal distribution is continuous. So it's a curve. Each column here is a different sample size.

This first column is sample size 5. Second column is sample size 10.

Third column 30 and 50. And then going down the rows, the true population parameter p changes.

Here it's 0.1, here it's 0.3. In the last row, it's 1 half. And so we see, if we look in this top left corner, sample size of 5 and true value of p equal to 0.1, so close to the boundary, right?

This normal approximation is not very good.

It doesn't really line up with the actual probability mass function. But we see, say, as p gets closer to the middle of the popular or the allowed values, right, 0.5.5. right in the middle of the 0-1 interval. Even with a small sample size here, sample size of 5, it does line up pretty well, pretty good approximation. Still doesn't account for the fact that this is discrete and that's continuous, but it approximates it pretty well. And we also see, going from left to right in any of these rows, as the sample size increases, the normal approximation gets better and better. It's pretty much perfect down here, right? But even for value of p of 0.1, close to the boundary, it's still a pretty good approximation. It falls pretty close to that normal approximation. And that happens because we can also see the spread of this is getting smaller and smaller. The variance is getting smaller and smaller. So even though P of 0.1 is close to the boundary, that variance is shrinking in enough that that boundary effect is getting smaller and smaller. I just wanted to point out some things here. We see as the sample size N increases, the sampling distribution becomes closer to a normal distribution. It's better and better approximated by a normal distribution. That's the central limit theorem. For p close to zero or one, I didn't show one here, but near the boundary, the sampling distribution we saw is skewed, right? And so that normal approximation isn't as great there. It approaches normality more slowly in this row, in this row. And as the sample size increases, we saw that sampling distribution has lower variance. we're learning more from our data. Yeah, question. Besides the true value first sample or do you sample that? We're going to talk kind of over the next couple of weeks of how we're going to use it. Some different ways we'll use it is to construct interval estimates. We need to understand the sampling distribution in order to do that or to make certain kind of statements, hypothetical statements about where our estimator might lie relative to the true value. So those are the types of things we'll do with it. Right now we don't really have those. tools yet, but we'll build up to those. It is the same table that I have in that first set of lecture notes. Okay. Yeah, question. Are these topics about sampling distribution? Are they going to be on the coming midterm or the one after? No. So today's lecture will not be on midterm one. It'll be covered on midterm two. Unless anyone has another question on this lecture, I can transition into the midterm review. So here, this is our probability mass function. The allowed values of T are 0, 1

over N, 2 over N through 1. So take the example, N is 5. The allowed values for T are 0, 1 5th, 2, 5th, 3 5th, 5th, 5th, 5th. That'll be the values on the x-axis. And the values on the y-axis will be this. So that's what we're plotting. This is the y-axis, the probability, but the value on the x-axis is the value realized. So in this case, we have 0, 1-5th, 2-5th, 3-5th, 4-fifth, 1. And then we have a probability associated with each of those values. So that's what it's showing. The vertical bars are literally the probability mass function. The blue curve is a normal approximation to that. Any other questions on this lecture? Just kind of going over kind of the main topics that will be covered on midterm one. Now, I'm not putting every detail in here, so make sure to go back over the first six lectures and readings and problems and everything. But just to kind of reinforce the basic ideas. So be very familiar with kind of these important definitions, right? We've talked about what is a population? It's a full set of units that we're interested in studying. Might be a finite population, might be an infinite population. But we don't typically get to observe that full population. So instead, we work with a sample or a subset of those units in the population, right? And we do get to observe measurements or data for those sampled units. Parameter is a numerical measurement that describes some characteristic of a population. This is typically the thing we would like to estimate or learn about. It could be a parameter of a specific probability distribution, but it doesn't necessarily have to be. Yeah, a question. No multiple choice questions. They're all solving problems similar to what you do in discussion section or homeworks. We've talked about frequentist inference and basing inference and the difference between those two. In the frequentest case, consider parameters to be some fixed but unknown value. They're not random. We don't think of them as random in the frequentist setting. In the Bayesian setting, we do think of them as random, or we treat them as random variables, at least. And so they have their own probability distribution, or maybe distributions giving, depending on when you observe the data. And so these distributions reflect your uncertainty or your state of knowledge about the distribution. And typically you have some state of knowledge before observing the data. We call that our prior distribution. And then you get to observe some data. You would update your beliefs. You've learned something about the parameter. And so then you get a posterior distribution. And we talked about the likelihood function, right? And here I've written down the likelihood function for a random sample, for IID data. It's a product of these probability mass functions or probability density functions depending on whether we have continuous or discrete data. And we view it typically as a function of the parameter condition on the data, at least in the frequentest case. And then we talked about baysian inference. I've got the prior distribution. And so if we have a statistical model with a parameter of theta, Treating theta as random, right, it has some distribution. And the distribution that we assigned to it before observing the data is our prior distribution. But then we get to observe some data X1 through XN. And so the distribution of theta conditional on those X's or given those X's is what we call the posterior distribution of the parameter. And Bay's theorem is how we actually get that posterior distribution. Bay's theorem says that that posterior is equal to the likelihood. these first N terms times the prior distribution on our parameter theta over this marginal distribution of the data. Or if we had written down the whole joint distribution of the parameter and the data and we integrate out the parameter, that gives us the marginal. And as we saw, we typically don't care about this marginal, right? We think of it as a normalizing constant that makes this integrate to one, so it is a valid distribution, but very often we instead say the posterior is proportional to the likelihood times the prior, and that's how we did it for the examples that we looked at. Yeah. Another term, should we be more concerned about, like, the posterior being proportional to, like, we've had times prior and then trying to worry about calculating calculating how to get a posterior distribution, right? If you have a prior distribution and you have a likelihood, can you give me a posterior distribution?

That's the important thing. We talked about conjugate priors, right? So for a given likelihood, if the prior and the posterior distribution belong to the same family of distributions, if they're both normal, or if they're both gamma, or they're both beta, right? If that's the case, we call that a conjugate prior

distribution. And when we have conjugate prior distributions, it makes things really nice, really convenient. You get these easy updating rules, right? We just update the parameters corresponding to that family of distributions based on the data that we have observed in some way.

And we looked at some important examples of that. So we talked about beta-Bernoulli conjugacy. So if we have a beta prior distribution for the parameter P associated with a Bernoulli data distribution, right, then we know that the posterior distribution will also be beta. The parameter alpha will be updated by adding the sum of the data X, and the parameter beta will be updated by adding the sample size N and subtracting the sum of the data X. We talked about gamma-poison conjugacy, right? So if we have a gamma-prior distribution for a parameter lambda associated with a Poison, data distribution, right?

Then we get conjugacy. The posterior is also going to be gamma, and the parameters are, again, simple updating rules. Alpha plus the sum of x, beta plus n. And then we talked about normal normal conjugacy. So if we have data, a normal distribution, a normal distribution is unknown, but the variance is known.

And we put a normal prior distribution on that unknown mean mu.

And that prior distribution will have its own mean and its own variance. In that case, again, we have conjugacy. We know that. the posterior distribution for mu will be normal, and we have these updating rules for the mean and the variance associated with that posterior. Now, Bayes theorem gives us an entire posterior distribution. It tells us kind of anything we might want to know about that distribution. Sometimes we just want a single best guess point estimate, right? And in the Bayesian setting, we use decision theory or use a loss function to get there. So we said a loss function is just some real value function of two variables in our case. L of Theta and A, theta is the parameter, true unknown parameter, and A is a real number or our guess. We interpret this as the loss that we realize, right, if the true parameter equals theta and our estimate equals A.

And we want to work with this posterior expected loss.

We don't know the loss itself, but because we have a distribution for theta, we can take an expectation or an average of this loss function.

And we know the definition of an expectation, right? It's just the integral of the function times the distribution. So we want to work with this posterior expected loss, right? And we said the Bayes estimator is the one that minimizes this posterior expected loss. And if we have this for observed data little X, we call that a base estimate. And we talked about two specific loss functions. The first was squared error loss, right? And so we saw that if we're using squared error loss, then the Bayes estimator or the one that minimizes the posterior expected loss is the posterior mean, the mean of the posterior distribution. If instead we were to use absolute error loss, then the Bayes estimator is the posterior median. If we have a symmetric distribution, we saw those are the same, but they're not necessarily always the same. Then we talked about maximum likelihood estimation or switching to a frequentist framework. So thinking of the parameter as fixed, but unknown, and we did that with maximum likelihood. And we defined the maximum likelihood estimator as the argument of theta or the input theta, that maximizes the likelihood function. And again, if we do this for observed or realized x values little x, we call that the maximum likelihood estimate of theta. So this is the definition. We always want to start here, but we saw we had this nice optimization procedure coming from calculus that can often help us find that MLE. We did this by first computing the log likelihood function because the log is monotonic, and it was easier to work with, products become sums and so on. And then we differentiate that log likelihood with respect to theta and set that equal to zero. We're finding a candidate point. Solve that expression for theta. That's our candidate MLE. We want that to be some maximum. And then we just check that it is indeed a maximum, check that

it's concave or that the second derivative is negative at the candidate point so that we're getting a maximum and not a minimum. But again, we should consider edge cases. We've saw plenty of examples where the MLE falls on the boundary of the parameters support.

Very often this comes up when we have the support of our data depending on the parameter, like that uniform example.

The parameter theta determines the allowable values of x. That's the MLE, but we talked about a couple of properties of the MLE as well. We have this nice invariance property where we said if we really are interested in this function G of theta, right, we can just plug in G of theta hat. We can get our MLE theta hat and plug that into the function G, and we get our MLE of this transformation of the parameter for free. And we talked about consistency, this asymptotic result. It says the MLE converges in probability to the true parameter. So those are the kind of main topics to go back over. But again, please go over lecture notes. The exam's going to be four questions, but they will have subparts. So, like A, B, C. You'll have the full class time to complete the exam. You can bring one single-sided sheet of handwritten notes. on standard printer-sized piece of paper but again that's single-sided and handwritten notes and you will turn that in when you turn in your exam no calculators you won't need them and they're not allowed as far as studying again it'll be pretty similar to homework or discussion section problems so it's good to review those problems also good to review the example problems I do in the notes also the couple kind of extra credit activity problems that we've done I think those are all great resources to go back over. If you understand all of those very well, I think you'll be in very good shape for the exam. You should know integration by parts, yeah. Can't say whether or not it will be on there. What are you thinking? You can have example problems, that's fine. Any other questions? Do you want to do another extra credit problem? Okay, so let's put everything. Electronics away, no cell phones, laptops, tablets. Take out a piece of paper and a pencil. Same as always, you can work together. Oh, and make sure to write your name on the paper. where I've had a couple turned in without a name on them. Okay, I will give you 20 minutes again starting now. You know what, did not have a location? I just wanted to have the best place to follow you. Okay. I was like looking at what structure will you do incorrectly in the process. And it was not approaching when you first. And I mean, the kind of general lesson of the past four lectures has been, go to me, go to me, go to me. Yeah. So we just filled over the audience. So we don't need, right? It's meo. Yeah. It's meo. Yeah. But we have to via view and go up, please. So we have a null mean, right? Yeah. But unknown. The deviation is just a square or variance. Yeah. Yeah, it's just signal. No, it's just a signal. So, guys, there's a formula. for the variance right yeah and then we use that to get the last question that's the standard the standard so we need to get the distribution for the first x-1 x-n right and then like that's like a no-brainer because we have the known mean and then we can like find the formula for the unknown variance and then we have the we have the distribution we're golden and then we just have to find the amylia of the synergy do you remember something like the variation Is the institution of x squared minus the unit, extradition borg? They write it down. Yeah, yeah, yeah, it's the institution the squared minus the x which is equal. Yeah, yeah, it's like an absolute. Yeah, yeah, yeah, yeah, it's the back. Yes, what's full? So, is the value? So it would be like the magnitude of this? It's like this, but this relates it with the hand. And then you do sort of that. But that's just the very good. Because I'm more of a close. You know what? Oh, I'm sorry. MLE of the standard deviation of this data. I think it's all. Let's just start with in order of lectures. Let's just go on distribution. Just to clarify, I'm not asking about the standard deviation of the distribution for the MLE, asking about the MLE of the standard deviation for the data distribution. Here we go, guys, I got it. Sorry, okay, so the variance, this is what I was trying to say, this, that's it, you know what I mean, the variance, sorry, I'm sorry, not the variance, the sigma square. Yeah, okay, you should vary. Hey dude, Mr. CEO, can you write this down? This is the one you need to write down. The sigma square equals that. Okay. Oh, sorry, yeah. So I just rid of the minus x bar and sub n. Yeah, and then I think it's just x bar, actually. And then the whole thing is

right. Yeah, that's right. Yeah, and that might not even be necessary, but the n might not even be necessary. Okay. It's just the notation thing. Anyway, what is that definition? That's the square. Cool. So we have a known mean, right? So we don't even need to guess about the meme. And then... Perimeter restriction. And we know that the mu is the sample mean. So in fact, we can just say that this is meep, this is me. Okay. In our case, because we have a known. Yeah, that's me. Thanks. I is. Yeah, you just take the square root of that. Yes, so it's speed, yeah. So it's just square root of that is the variance. Okay, sorry. Squared of sigma, of sigma squared is sigma, which is. That's right? Yeah, that's the standard. Okay, I never know. Sorry. It's just right. Yeah, yeah, yeah. Okay, pull it up, thank you. Right here. Yeah, so just one second. Like, give me like, 45 seconds. So, like, you see that equation you wrote up there, right? The second one, right? Yeah, the variance. If you take the square root of that, that's just the standard deviation. And then we already know the means. And then we already know the means. so you can just replace that x bar end with the what are we trying to find right now we're trying to find the ammo of the same here I got that I'm saying what precisely in this process are we trying to look at it is that the standard standard variance okay okay I'm sorry I'm sorry you need a hat you need a hat or is a hat on top of sequence right yeah because that's the maximum estimate I mean the whole thing The whole sigma square. Where do you get that equation from? No, the hat goes on top of the whole thing. The hat goes here on top of the term. It's just like, I think he just meant like, it's right here, and then we said that, yeah, I think you're just going to be the summation of XI, so. Yeah, yeah, and that's going to be new. Yeah, and that's going to be new because, But it's a sample with the end of the definition of talking about. Yeah. And then he just, where we're going to point, he wasn't needed the sample mode to say. Yeah, I'm confused about that. He got that as the definition of six squared. You said you were going to show me like what? Do you think, uh, I don't think I've seen this. I mean, I know this is correct just because of like, Econ 131 or Egon, it doesn't matter. That is correct. I just don't know if there are a page that changes that definition, you know what I mean? So, let's say most right is the very. I'm not. Oh, like, it's strong, like, this is it's not. It's like, is like, You should try to find the mic and good. Okay. It's not. Yeah. 10 minutes and 10 minutes. 10 minutes. 10 minutes. 10 minutes. I don't know where they have minus one. Yeah, I'm just worried about that. She basically derived it as a traditional way of playing the like computer function. There could be something for longer with like $100 and $10. So the action is big? It should be. I'm doing it. I'm doing it the long. And then you actually did like the, yeah. I didn't all thousand. Oh, wow. That is, yeah. Thank you. Thank you. Raise it. Raise it. You just raise the... You just raised the... ...that you find the different language by people that you're giving in respect to the... Oh, you do you want to put the closet? Yeah. Do you want to quit the question? She might be a good thing. Yeah, yeah. What don't you get? Yeah. Okay. Okay. There's no one who explained the class, so we got the answer. Oh, yeah, that's supposed to be in red. To be clear, this is my calculation. I cannot speak for how correct it is, but we know that we have the distribution. I'm just basically rewriting. There is already there. I basically approached this in the way that we solved for the emily in class. Which is the first to find the likelihood, which is the sum of all of that. Sorry, product of all of that. Here's a constant, so we can just take that to the power of n. Sorry, I also my handwriting sucks. And then within this exponent, we have the sum from one to n of sine like to sigma square over to sigma square. You can take the log. If you know your log rules, that's great. If it's, if you don't. Um, and then here this just, what I did now is then take derivative with respect to sigma square was, I think, kind of end to the two plus two. It's just like calculus, so it's hard to explain that. right now but excuse me I think the summation should be inside of the premises and for x-i and so it should be where the summation should be inside of the square x-i you know right okay and then let this equal to zero and then you said it equal to zero this because sorry I'm skipping stuff I multiplied both sides by two by this So that gets rid of that here and you can end up with. And since this is, we want this. So that hits the square root. I'm sorry, is that one of the theory you checked out too. I just thought I just don't know how else. I don't want to give anything away. That's how I answered everything. Oh, right, that's my area of the section. Start turning in, folks. They really suck you if we all dropped

it or that's all. Just on the answer. Just on the answer. Yeah, I don't need the work at all. Okay, two minutes. Okay, do you know my husband? Okay, he's my husband. Yeah, we're really good down. Well, I might get those. It turned out we're wrong. We're all wrong. This is the front of math, you know? You're very certain about very little. Does anybody else need to turn theirs in? Okay, last call. Okay, so the answer was B. Still going to go over it real quickly, so it'll be captured on the lecture feed, and you can watch it when you study later. Yeah, so step one, we need to write down the likelihood function. We need that if we're going to maximize it. So we want the likelihood function for sigma squared, give them the data X, and because we have a random sample, it's going to be a product of these N normal probability density functions. 2 pi, sigma squared, and negative 1.4. half e the negative x i minus mu squared as a terrible squared squared squared over two sigma squared okay collecting terms simplifying we have two pi sigma squared and minus n over two e the negative summation overall i x i minus mu squared two sigma squared okay that's our likelihood function we can take the log to make our lives easier a little l sigma squared the x i is going to be equal to constant minus n over two log squared minus summation x i minus mu squared for two sigma squared okay and then we want to differentiate this log likelihood function respect to our unknown parameter sigma squared it's going to give us minus n sigma squared had over that the summation x i minus mu squared over two squared squared squared and we want to to set this equal to zero and solve for our candidate, sigma squared hat of n to a squared hat is equal to the summation xI minus mu squared to hat squared squared and then isolating the squared hat we get summation of xI minus mu squared over n. Okay, but that's for, that's our MLE for sigma squared. The variance, we actually want the MLE for the standard deviation. But we know the standard deviation sigma is the square root of sigma squared. Sigma hat MLE of the standard deviation is the square root of the MLE for the variance or the square root summation xI minus mu squared. So we did two things here. We got the maximum likelihood estimate for the variance. And then we used our invariance property of the MLE to get the MLE for the standard deviation. There's often multiple ways to go about a problem. you could have also found the MLE or Sigma directly the likelihood function in terms of sigma, differentiate with respect to Sigma, and you'd get to the same answer.

Either way it would be fine. Say that again? So if you look on the really short-hand reset process to find the answer where we immediately are able to... So you're jumping from here to here? No, actually there was no jump. That's not connected. It's the Sigma, Sigma Square hat just from there down.

Just from the fact that we have the view and then we know that that sigma square hat is that and then we plug in the mu and then we're... That's just, like, it's from the textbook. The formula for a sigma squared, the hat was pulled from the textbook and then just take the square to try. Actually, it's kind of me assuming that you can do the mean, you have a known mean and unknown variance and then you use that formula. Is that any correct assumption? So I don't know what specific formula you were looking at. Was it for the NLE? When you do a normal, and then there's, with unknown, mean and unknown variance, right? So then I just assumed that the formula for the variance was the same. And then I took that mean, which is I plugged it into the variance. Yeah, I guess what I'm getting at is there's multiple different estimators for the variance. The MLE is one possible estimator for the variance. So it would depend on which formula you're using and why you're using it. Yeah, if it's for the MLE, then, yeah. So for these extra credit problems, you can use the textbook. That's fine. You wouldn't have it on the exam, and I would want you to show your work, and citing the textbook would not be showing the work on the examines. Unknown variance, normal distribution, right? And then you're in the maximum likelihood, yeah. And then you go down here, and then it says, okay, so the mean is the sample mean, and then the variance is this, right? And so if I said this is the variance, then this is the formula for the variance, given that we know what mu is, right, because we have a no mu. I just plug the mu in there, and then we go forward to that. Yeah, but you skip this step. which is basically what I was doing. And it's the same thing. Yeah, but it is like a, I mean, like it's another way to get to the same thing, given the problem. Yeah, that's generally if you're, if

you want the MLE for both when they're both unknown. But yeah, again, on exam, I'd want you to show something like this. Yeah, because I, again, any theorems I've given you in the notes, you can use those theorems on the exam, but you can't just take any theorem you find anywhere for the exam. Even if it's like, section, too. This thing about the unknown variance and unknown mean was also in section. Don't we have a sure, like a class about... I don't remember any of the variance, though. We have the joint distribution of the mean and the variance next week, or no, the week after. Maybe it was... Okay, so you want us to do it that way. Yeah, again, feel free. If I've given you a theorem in the notes, have that theorem on your note sheet, you can use it directly. But if it wasn't a theorem, I gave you, and I asked you to... Like, if I ask you to do a problem that you had done on the homework, I would still want you to do the problem on the exam. I wouldn't want you to say, here's the answer, right? All right. I'll stick around if there's any questions. Otherwise, have a nice weekend, and good luck on the exam. Tuesday.